

FINAL PROJECT REPORT

Import Data Using Transform Maps

Importing employee data from a spreadsheet into ServiceNow using Import Sets and Transform Maps

Date	28 September 2026
Team ID	SWTID-2026-6491
Project Name	Import Data Using Transform Maps

Table of Contents

1. Introduction

- 1.1 Project Overview
- 1.2 Purpose

2. Ideation Phase

- 2.1 Problem Statement
- 2.2 Empathy Map Canvas
- 2.3 Brainstorming

3. Requirement Analysis

- 3.1 Customer Journey Map
- 3.2 Solution Requirement
- 3.3 Data Flow Diagram
- 3.4 Technology Stack

4. Project Design

- 4.1 Problem – Solution Fit
- 4.2 Proposed Solution
- 4.3 Solution Architecture

5. Project Planning & Scheduling

- 5.1 Project Planning

6. Functional and Performance Testing

- 6.1 Performance Testing

7. Results

- 7.1 Output Screenshots

8. Advantages & Disadvantages

9. Conclusion

10. Future Scope

11. Appendix

1. Introduction

1.1 Project Overview

This project imports bulk employee data received as an Excel spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The spreadsheet is first loaded into an Import Set table (Employee Importing, u_employee_importing), which acts as a staging area. A Transform Map named “Sample Spreadsheet Import” then maps the source fields to a custom target table, Employee Test (u_employee_test), and creates the records on execution. The sample spreadsheet (Employee_proj.xlsx) contains 15 employee records with the columns Employee ID, Name, Email ID, Department and Location.

The solution is extended with Coalesce on the Employee ID field so that repeated imports update existing employees instead of creating duplicates, and with three ServiceNow reports (Employees by Department, Employees by Location, Employee List) that are consolidated into the Employee Analytics Dashboards.

1.2 Purpose

Employee data usually arrives from external sources as spreadsheets and must be moved into the target system accurately and without duplicates. Entering such data by hand is slow and error-prone. The purpose of this project is to support structured data import, accurate field mapping and transformation, and validation of the resulting records, so that the employee data in ServiceNow stays clean, consistent and ready for reporting even when the same data is imported several times.

2. Ideation Phase

2.1 Problem Statement

Imported source data may require field mapping and transformation before it can be used correctly in the target system. Using the “I am / I’m trying to / but / because / which makes me feel” framework, three problem statements were defined for the people affected by manual employee-data handling:

PS	I am	I’m trying to	But	Because	Which makes me feel
PS-1	an HR executive receiving employee lists in Excel	load the whole list into ServiceNow quickly	I have to re-type or copy records one by one	there is no automated way to map spreadsheet columns to the system fields	overloaded and worried about typing mistakes
PS-2	a ServiceNow administrator	import the same employee file again when details change	duplicate employee records get created	existing records are not matched on a unique key such as Employee ID	frustrated and unsure the data can be trusted
PS-3	an HR manager or analyst	see how employees are spread across departments and locations	I need to count and sort the data manually	no ready reports or dashboard exist on the imported data	uninformed and slow to make decisions

2.2 Empathy Map Canvas

To understand the user's experience before the solution, an empathy map was prepared for the persona "HR Executive Importing Bulk Employee Data".

Quadrant	HR Executive importing employee data
Says	"I receive the employee list in Excel every time." "Please don't create the same employee twice." "Can I see the result quickly?"
Thinks	Will the columns match the right fields? What if I upload the same file again? Is the data safe and correct once it is in the system?
Does	Prepares the sheet, downloads it as .xlsx, uploads it through Load Data, checks the imported list and shares numbers with the team.
Feels	Anxious about mismatched fields and duplicates; relieved when the import shows Completed and Success.
Pains	Manual typing, wrong or missing field mapping, duplicate records on re-import, no quick summary of the data.
Gains	Fast bulk import, reusable field mapping, no duplicates through Coalesce, ready reports and a dashboard for HR.

Table 2.1: Empathy Map – HR Executive Importing Bulk Employee Data

2.3 Brainstorming

The team explored approaches for reliable data import, field mapping, transformation, validation and error handling, and grouped the ideas into four themes:

- **Cluster A – Import Set & Transform Map (selected):** import the spreadsheet through an Import Set table and a Transform Map; map fields with Auto Map and Mapping Assist; run the transform and check the transform history.
- **Cluster B – Data Quality & Duplicate Control:** use Coalesce on Employee ID so repeated imports update records; review inserted, updated and ignored counts; correct field maps when a field is not mapped.
- **Cluster C – Reporting & Dashboards:** build reports by department and location, a list report, and combine them in one dashboard.
- **Cluster D – Automation & Integration (future scope):** scheduled imports, integration with LDAP/REST sources, automatic error notifications.

Clusters A and B directly solve the problem statements PS-1 and PS-2 and are feasible with standard ServiceNow features and no custom scripting, so they form the core of the project. Cluster C was added as an extension to answer PS-3, while Cluster D was kept for future work.

3. Requirement Analysis

3.1 Customer Journey Map

The journey of preparing source data, importing, transforming, validating and reviewing the resulting records was mapped across five phases to show friction points and opportunities.

	Entice	Enter	Engage	Exit	Extend
Stage	Employee data arrives from HR as a spreadsheet	User opens ServiceNow and starts Load Data	Data is loaded, mapped and transformed	Records are checked in the target table	Data is re-imported and analysed
Actions	Prepare the sheet and save it as .xlsx	Choose the import set table, file, sheet 1 and header row 1	Auto Map, Mapping Assist, click Transform	Open Employee Test and review the list	Enable Coalesce, load updated file, build reports
Pain points	Inconsistent column names	Wrong table or file selected	Fields not mapped, transform fails	Missing or wrong values	Duplicates on every re-import
Opportunities	Use a fixed sheet format	Guided Load Data form	Reusable Transform Map	Transform history and list view	Coalesce, reports and dashboard

Table 3.1: Customer Journey Map – Spreadsheet Import into ServiceNow

3.2 Solution Requirement

Functional Requirements:

FR No.	Epic	Sub Requirements
FR-1	Source Data Preparation & Load	Prepare the spreadsheet and download as .xlsx; load it through Load Data into the Employee Importing table (sheet 1, header row 1).
FR-2	Target Table Creation	Create the Employee Test table (u_employee_test) with String fields: Employee ID, Employee Name, Email, Department, Location.
FR-3	Transform Map & Field Mapping	Create the Sample Spreadsheet Import map from Employee Importing to Employee Test; use Auto Map Matching Fields and Mapping Assist; run Transform.
FR-4	Duplicate Prevention	Set Coalesce true on the Employee ID field map so repeated imports update existing records.
FR-5	Validation of Results	Verify records in Employee Test and the inserted / updated / ignored counts in the transform history.
FR-6	Reports & Dashboard	Create the pie, bar and list reports and add them to the Employee Analytics Dashboards.

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Simple Load Data and Transform Map screens; clear success messages after import and transform.

NFR No.	Requirement	Description
NFR-2	Security	Only users with admin access configure tables, transform maps and dashboards; access can be limited by users, groups and roles when sharing reports.
NFR-3	Reliability	The same Transform Map gives the same result each time; re-importing the same file causes no duplicate records.
NFR-4	Performance	Import and transformation of the project dataset complete within seconds.
NFR-5	Availability	Runs on the ServiceNow platform and inherits its availability; no extra server is needed.
NFR-6	Scalability	The Import Set table and Transform Map can be reused for larger files and later imports.

3.3 Data Flow Diagram

The Level-1 DFD below traces how data moves from the spreadsheet to the target table and on to reports:

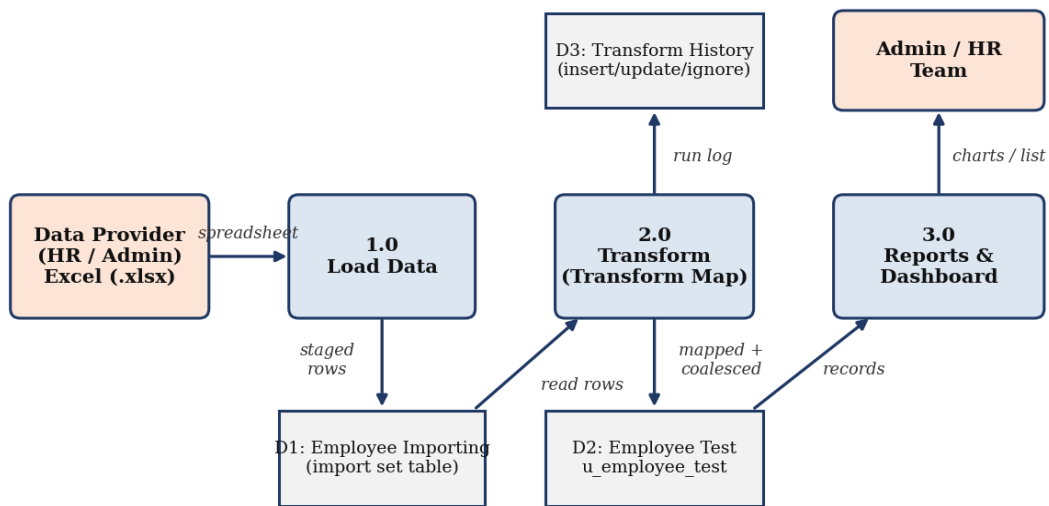


Figure 3.1: Level-1 Data Flow Diagram

3.4 Technology Stack

The solution is built entirely on native ServiceNow components, with Google Sheets / Excel used only to prepare the source file:

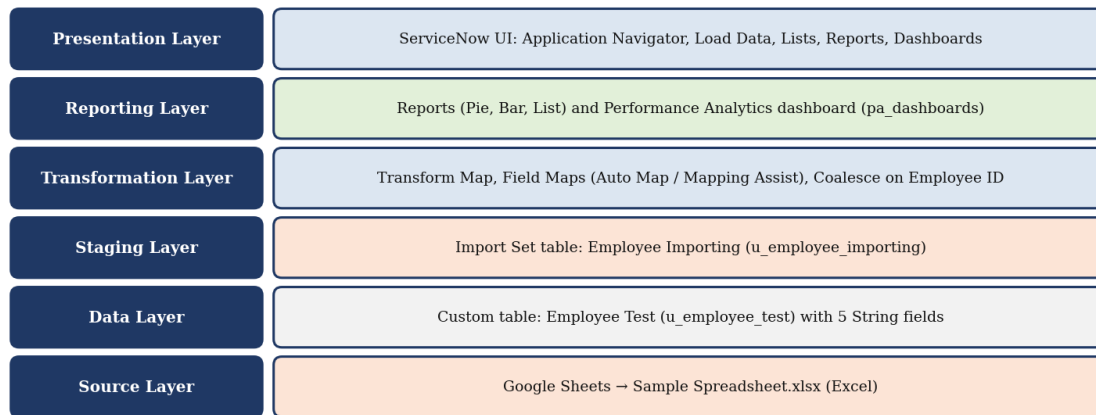


Figure 3.2: Layered Technology Stack

4. Project Design

4.1 Problem – Solution Fit

The problem-solution fit confirms that the proposed import process matches the identified need:

- **Customer Segment:** HR executives and ServiceNow administrators who receive employee data as spreadsheets.
- **Jobs-to-be-Done:** get employee data into ServiceNow quickly, correctly and without duplicates.
- **Root Cause:** source columns are not automatically linked to target fields, and repeated imports have no unique key to match on.
- **Triggers:** a new employee list, an updated employee file, or a request for department and location numbers.
- **Your Solution:** an Import Set and Transform Map with field mapping, Coalesce on Employee ID, and reports on the imported data.

4.2 Proposed Solution

Problem Statement	Imported source data needs field mapping and transformation before it can be used correctly in the target system, and repeated imports can create duplicates.
Idea / Solution	Configure an import process and Transform Map with appropriate field mappings, Coalesce on Employee ID, and reports and a dashboard on the result.
Novelty	Uses standard ServiceNow features only (Load Data, Import Set, Transform Map, Reports) with no custom scripting, and adds Coalesce so the import can be run repeatedly.
Social Impact	Less manual data entry, more accurate employee records and quicker HR decisions.
Business Model	Internal efficiency tool; value comes from saved manual effort and cleaner data rather than revenue.
Scalability	The same Import Set and Transform Map pattern can be reused for other tables and larger datasets.

4.3 Solution Architecture

Source data → Import Set → Transform Map → Field Mapping / Transformation → Target Table. The end-to-end architecture is shown below:

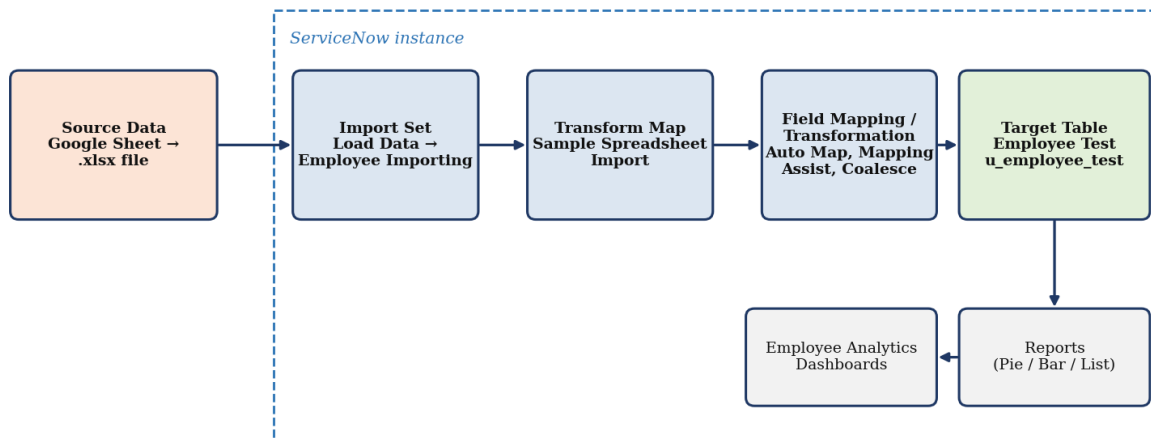


Figure 4.1: Solution Architecture – Import Data Using Transform Maps

The source spreadsheet columns are mapped to the target table fields as follows:

Source column (Employee Importing)	Target field (Employee Test)	Note
Employee ID	Employee ID	Coalesce field – unique key used to avoid duplicates
Name	Employee Name	Mapped with Auto Map / Mapping Assist
Email ID	Email	Mapped manually with Mapping Assist (see BG-002)
Department	Department	Values: ServiceNow, Salesforce
Location	Location	Value: Hyderabad

5. Project Planning & Scheduling

5.1 Project Planning

Activities included source-data preparation, Transform Map configuration, testing, validation, documentation and final demonstration. The backlog of 10 user stories was estimated at 30 story points and scheduled across two sprints, followed by User Acceptance Testing.

Story	User Story	Points	Sprint	Planned Dates
US-1	Prepare the employee spreadsheet and download it as .xlsx	2	Sprint-1	08 Sep – 13 Sep 2026

Story	User Story	Points	Sprint	Planned Dates
US-2	Create the Employee Test table with 5 String fields	3	Sprint-1	08 Sep – 13 Sep 2026
US-3	Create the Employee Importing import set table through Load Data	3	Sprint-1	08 Sep – 13 Sep 2026
US-4	Create the Transform Map and Auto Map matching fields	5	Sprint-1	08 Sep – 13 Sep 2026
US-5	Map fields with Mapping Assist and run Transform	3	Sprint-1	08 Sep – 13 Sep 2026
US-6	Validate the created records in Employee Test	2	Sprint-2	14 Sep – 19 Sep 2026
US-7	Enable Coalesce on Employee ID	3	Sprint-2	14 Sep – 19 Sep 2026
US-8	Re-import updated data and check inserted / updated / ignored counts	3	Sprint-2	14 Sep – 19 Sep 2026
US-9	Create the three reports (Department, Location, List)	3	Sprint-2	14 Sep – 19 Sep 2026
US-10	Create the dashboard and add the reports to it	3	Sprint-2	14 Sep – 19 Sep 2026

Sprint	Focus	Story Points	Duration	Planned Dates
Sprint-1	Source data, custom table, import set, Transform Map, field mapping and first transform	16	6 Days	08 Sep – 13 Sep 2026
Sprint-2	Validation, Coalesce, re-import, reports and dashboard	14	6 Days	14 Sep – 19 Sep 2026
UAT	User Acceptance Testing and report submission	–	3 Days	20 Sep – 22 Sep 2026

Average Velocity (Sprint-1) = 16 story points ÷ 6 days $\approx$ 2.7 points/day. The ideal burndown for Sprint-1:

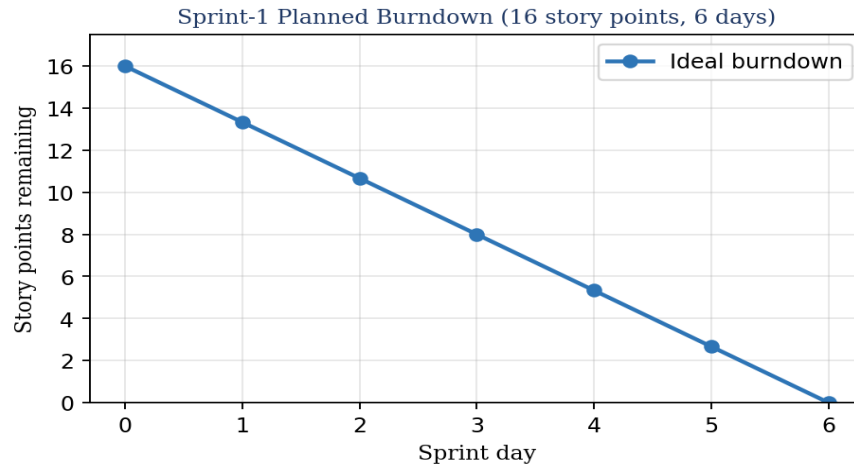


Figure 5.1: Sprint-1 Planned Burndown

6. Functional and Performance Testing

6.1 Performance Testing

User Acceptance Testing was executed on 22 September 2026. Four test cases cover the import, the Transform Map, the field mapping and the data validation including Coalesce behaviour. All four passed.

Test Case ID	Scenario	Actual Result	Section	Result
TC-001	Import spreadsheet data into the Import Set table	Load completed with State Complete and Completion code Success: 11 rows processed, 11 inserts, 0 updates, 0 errors, in the Employee Importing table.	Import Data	Pass
TC-002	Create and configure the Transform Map	Transform Map “Sample Spreadsheet Import” created and saved.	Transform Map	Pass
TC-003	Verify field mapping between source and target tables	All five fields mapped correctly between Employee Importing and Employee Test.	Field Mapping	Pass
TC-004	Validate transformed data and Coalesce behaviour	First import inserted the records correctly; on re-import 2 records updated and 2 inserted; the same file loaded again gave 0 inserted, 0 updated, 4 ignored.	Data Validation	Pass

Two defects were found during testing and both were fixed and closed:

Bug ID	Description	Severity	Status	Resolution
BG-001	Duplicate employee records created when the same spreadsheet was re-imported.	Medium	Closed	Coalesce set to true on the Employee ID field map.

Bug ID	Description	Severity	Status	Resolution
BG-002	Auto Map Matching Fields left the Email field unmapped.	Low	Closed	Email mapped manually with Mapping Assist and the map saved.

Test case analysis: 4 total, 0 not tested, 0 failed, 4 passed. Defect analysis: 2 defects, both fixed. The transform history of the re-import run showed 4 rows uploaded, of which 2 were inserted (new Employee IDs) and 2 updated (changed name and email). Full test steps and the bug log are maintained in the separate UAT document and UAT Execution Report.

7. Results

7.1 Output Screenshots

The spreadsheet data was loaded into the Employee Importing staging table, transformed into the Employee Test table through the Transform Map, and reports were built on Employee Test. The first load completed with State Complete and Completion code Success (11 rows processed, 11 inserts, 0 updates, 0 errors). Values shown by the reports on the dashboard:

Report	Result shown on the dashboard
Employees by Department (Pie)	Salesforce – 12 (46.15%), ServiceNow – 12 (46.15%), (empty) – 2 (7.69%)
Employees by Location (Bar)	Hyderabad – 24, (empty) – 2
Employee List (List)	26 records in total (page 1 to 20 of 26) with Employee ID, Name, Email, Department, Location

Screenshot :

The screenshot shows the 'Load Data' interface in ServiceNow. The browser address bar displays 'dev395351.service-now.com/now/ui/classic/params/target/create_import_set.do'. The interface includes a 'Load Data' button and a 'Submit' button. The 'Import set table' section has two radio buttons: 'Create table' (selected) and 'Existing table'. Below this, the 'Label' is 'Employee Importing' and the 'Name' is 'u_employee_importing'. The 'Source of the import' section has two radio buttons: 'File' (selected) and 'Data source'. Under 'File', there is a 'Choose File' button and a text field showing 'No file chosen'. Below this, the 'Sheet number' is '1' and the 'Header row' is '1'. A 'Submit' button is located at the bottom of the form.

dev395351.servicenow.com/now/nav/ui/classic/params/target/create_import_set.do

servicenow All Favorites History Workspaces Load Data Search Submit

Load Data

Import set table ☐ Create table ☒ Existing table

Import set table Employee Importing [u_employee_importing]

Source of the import ☒ File ☐ Data source

File Choose File No file chosen

Sheet number 1

Header row 1

Submit

dev395351.servicenow.com/now/platform-analytics-workspace/pages/params/library/dashboards

servicenow All Favorites History Workspaces Platform Analytics Search Create dashboard

Overview Libraries Dashboards Data visualizations Indicators Filters Scheduled exports Additional Links Usage insights

Dashboards

Create dashboard

Updated in the las... 2

Active Views

Analytics Usage Overview True

Application Services Dashboard True

New dashboard

Select how you want to create your dashboard.

☒ In-line editor Recommended ☐ Technical editor

Name Employees by Location

Description

Create dashboard Cancel Create dashboard

dev395351.servicenow.com/now/nav/ui/classic/params/target/import_status.do%3Fsysparm_ds_sysid%3D510c3780472b43104b8db3f4116d43d9%26sysparm_create_t...

servicenow All Favorites History Workspaces ServiceNow Search

Progress

Name ImportProcessor

State Complete

Completion code Success

Message Processed: 11, inserts 11, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:00.615)

Next steps...

Import sets Go to the import sets for this data load

Loaded data Go to the newly imported data inside the staging table: u_employee_importing

Create transform map Create a transform map for the newly staged data

Run Transform Transform a loaded import set using an existing transform map

Import log View the import log

servicenow All Favorites History Workspaces | Employee List Report Search

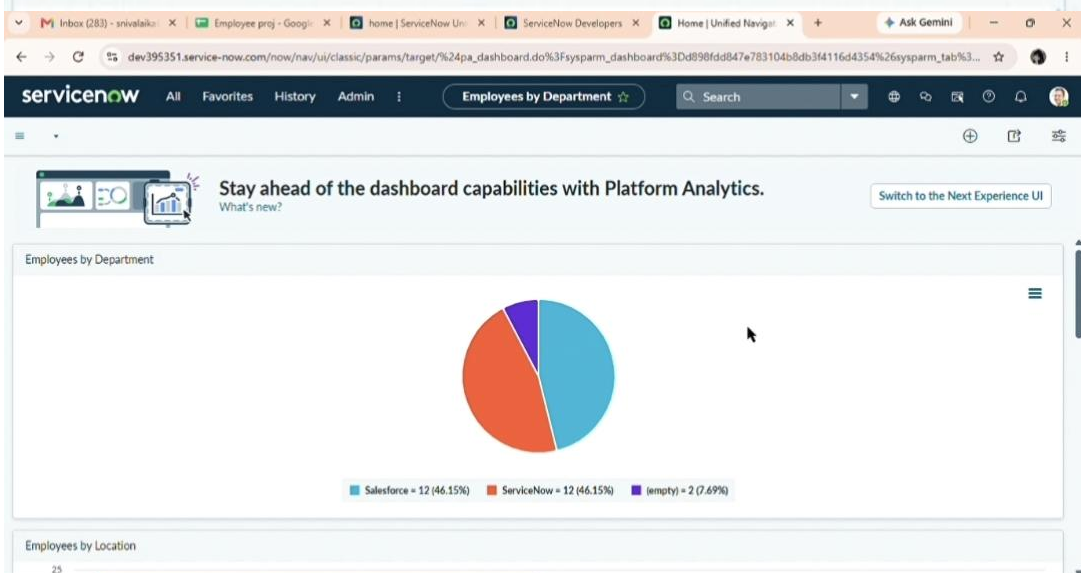
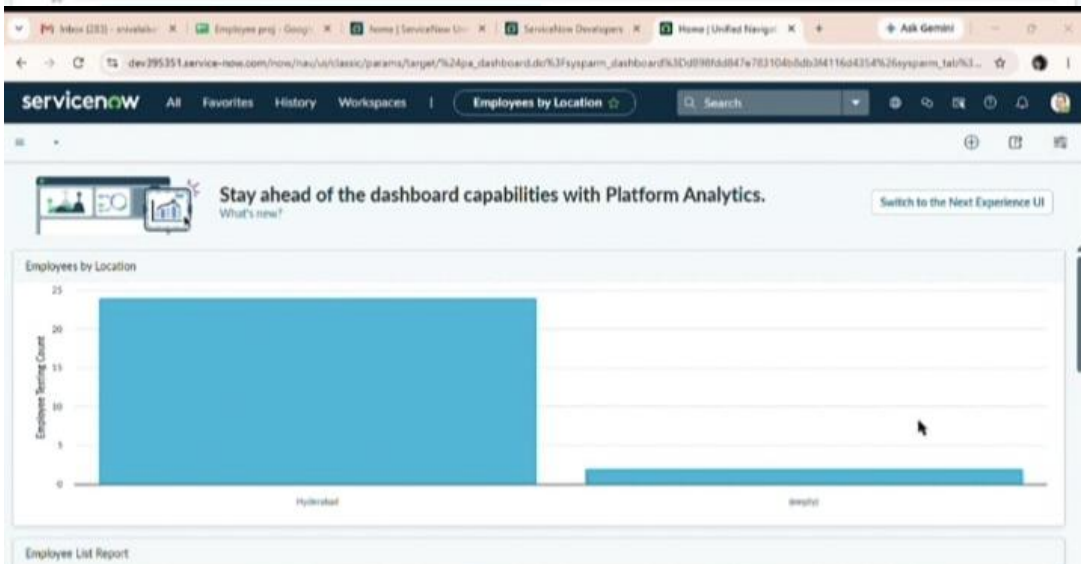
Stay ahead of the dashboard capabilities with Platform Analytics. What's new? Switch to the Next Experience UI

Employee List Report

Employee ID	Name	Email	Department	Location
SB-0009	Uka Kalai	test9@gmail.com	Salesforce	Hyderabad
SB-0001	Rattan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0010	Vincy Kumar	test10@gmail.com	ServiceNow	Hyderabad
SB-0006	Nithya Vani	test6@gmail.com	Salesforce	Hyderabad
SB-0002	Sandree Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandree Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0006	Nithya Vani	test6@gmail.com	Salesforce	Hyderabad

1 to 20 of 26

Employees by Location



8. Advantages & Disadvantages

Advantages:

- Structured data transformation from spreadsheet to target table.
- Reusable field mappings and Transform Map for later imports.
- Coalesce prevents duplicate records and updates existing ones.
- Improved consistency of imported records and less manual typing.
- Reports and a dashboard give quick visibility of the employee data.

Disadvantages:

- Incorrect mappings or transformation logic can result in invalid or incomplete target data.
- The source spreadsheet must keep the same column structure.
- Creating dashboards through the backend form needs admin access to the pa_dashboards ACL.

9. Conclusion

The Import Data Using Transform Maps project demonstrates an approach for importing source data and transforming it into the required target structure. Employee data from a spreadsheet was loaded into an Import Set, mapped and transformed into the Employee Test table, and validated through UAT with all test cases passing.

Enabling Coalesce on Employee ID made repeated imports safe: existing employees are updated and only new Employee IDs are inserted, which keeps the data clean and consistent. Reports and the Employee Analytics Dashboards give HR a single view of employees by department and location.

10. Future Scope

- Enhanced validation rules before records are created.
- More complex transformation logic, such as scripts and field-value conversions.
- Automated error reporting and email notifications after each import.
- Scheduled imports and support for larger datasets and other sources such as LDAP or REST.

11. Appendix

Source Code (if any):

No custom code was written; the solution uses configuration only. Key configuration details:

Target table	Employee Test (u_employee_test)
Target fields	Employee ID, Employee Name, Email, Department, Location (all String)
Import Set table	Employee Importing (u_employee_importing)
Transform Map	Sample Spreadsheet Import (Employee Importing → Employee Test)
Coalesce field	Employee ID

Reports	Employees by Department (Pie), Employees by Location (Bar), Employee List (List)
Dashboard	Employee Analytics Dashboards

Dataset (Employee_proj.xlsx – Sheet1, 15 records):

Employee ID	Name	Email ID	Department	Location
SB-0001	Rattan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandree Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Lavanya Deep	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Essai Manoj	test4@gmail.com	Salesforce	Hyderabad
SB-0005	Aki Tharun	test5@gmail.com	Salesforce	Hyderabad
SB-0006	Nithiya Vani	test6@gmail.com	Salesforce	Hyderabad
SB-0007	Kaviya Mari	test7@gmail.com	Salesforce	Hyderabad
SB-0008	Prekash Kiven	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Lisa Kalai	test9@gmail.com	Salesforce	Hyderabad
SB-0010	Lincy Kumar	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Pusgalu Devid	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Selvi Sir	test12@gmail.com	ServiceNow	Hyderabad
SB-0013	Grace Deviya	test13@gmail.com	ServiceNow	Hyderabad
SB-0014	Akila Perusu	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Jikil Preeven	test15@gmail.com	ServiceNow	Hyderabad

Dataset Link:

<https://docs.google.com/spreadsheets/d/1RDmv22mbHeBuqR7nThSiBXoeOuvogQpR/edit?usp=drivesdk&ouid=109528725825762463316&rtpof=true&sd=true>

GitHub Link: <https://github.com/sudharrajan7-cpu/Import-Data-using-Transform-Maps-Spreadsheet-.git>

Project Demo Link: <https://drive.google.com/file/d/11DkzMLE-RBx-BZ4tgGSRw-gktqOOMtxR/view?usp=drivesdk>